A Two-Node Counterexample to Naive Typed Patchwork

Witness-menu propagation is load-bearing for $\mathcal{ALCI}_{\mathsf{RCC5}}$

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Disclaimer. This document was generated by Claude (Anthropic), prompted by Michael Wessel. It corrects a conjecture made in an earlier probe paper by the same author [1]. The results have not been independently verified. We invite scrutiny and corrections.

Abstract

In an earlier probe paper [1] we conjectured (Proposition 7.1) that arc-consistent typed RCC5 networks with non-empty filtered domains are always realisable by some model of the $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox. The conjecture, under the weak form of arc-consistency used in that paper (composition plus type-safety propagation over existing nodes only), is *false*. We exhibit a two-node counterexample: a TBox with seven axioms under which the network $\{x, z\}$ with natural Hintikka types is arc-consistent but has no model, because a forced $\exists\mathsf{PP}.D$ witness at x cannot be related to z by any atomic relation compatible with z's universals. The failure is local in the network but non-local in the TBox: it sits in the gap between "every edge of V is type-safe" and "every pending existential at V is extensibly type-safe." We analyse why the counterexample works, propose the minimal correction (adding witness-menu propagation to arc-consistency), and observe that the corrected conjecture essentially reconstructs the split-forest paper's rank-d witness-menu machinery [2]. The correction does not affect the decidability claim for $\mathcal{ALCI}_{\mathsf{RCC5}}$; it does however sharpen the structural picture and explains why split-forest's apparent complexity is load-bearing rather than cosmetic.

1 Introduction

The patchwork-augmentation probe paper [1] investigated whether typed arc-consistency is complete for realisability of $\mathcal{ALCI}_{\mathsf{RCC5}}$ typed networks. Three natural counterexample probes were attempted; each collapsed to local arc-inconsistency or type-level unsatisfiability, suggesting that naive typed patchwork might hold. The paper conjectured this but did not prove it (Proposition 7.1, stated conjecturally).

This paper provides the counterexample we missed. It is smaller than anything we tried: just two nodes, seven TBox axioms, no triangles. The failure mode we overlooked was not a failure at any arc of the existing network, but a failure to *extend* the network with a pending

existential witness. Arc-consistency on V, as we defined it, does not look past V; so a TBox that forces a witness incompatible with some $z \in V$ is not caught.

The correction is conceptually minor—add existential propagation to arc-consistency—but the consequence is structural: the corrected propagation is exactly what the split-forest paper's rank-d witness-menus compute. The simpler patchwork-style argument we were hoping for in [1] does not exist. Any honest decidability proof for $\mathcal{ALCI}_{\mathsf{RCC5}}$ has to do the witness-menu work, either directly or via an equivalent mechanism.

What this does and does not say. This paper corrects a specific proposition from a specific probe paper. It does *not*:

- Falsify the decidability of $\mathcal{ALCI}_{\mathsf{RCC5}}$. The split-forest soundness theorem [2] and the completeness extraction theorem [3] remain intact; neither relied on naive arc-consistency.
- Falsify any of the three probes in [1]. Those still reduce to cheap local obstructions under the corrected definition of arc-consistency too; the issue is elsewhere.
- Affect the cover-tree tableau implementation [4]. CT1–CT5 are a different propagation scheme and do not use naive arc-consistency either.

What it does say: the probe paper's sketch of "a simpler alternative decidability proof via arity-≤ 3 CSP completeness" ([1], §9) is too optimistic. Witness extensibility is a genuinely harder constraint than pairwise type-safety, and it must be handled explicitly.

2 The weak form of arc-consistency

We recall the setup from [1], using identical notation. Fix an $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox T and concept C_0.

Definition 2.1 (Weak arc-consistency; [1], Def. 4.3)**.** A *disjunctive typed network* is (V, D, τ) with V finite, $D : V \times V \to 2^{\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}}$ satisfying inv-symmetry, and $\tau : V \to \mathrm{Tp}(C_0, T)$. Weak arc-consistency is the fixed point of

$$D'(x, z) := D(x, z) \cap \bigcup_{R_1 \in D(x,y)} \mathrm{comp}(R_1, D(y, z)) \cap \mathrm{TypeSafe}(\tau(x), \tau(z))$$

over all $x, y, z \in V$. A network is *weakly arc-consistent with non-empty domains* if the fixed point has $D(x, y) \neq \varnothing$ for every pair.

This propagation uses only (i) the RCC5 composition table on triples within V and (ii) pairwise type-safety. Crucially, it ignores the existential content $\exists R.D \in \tau(x)$: obligations to *extend* the network are not propagated back to constraints on existing nodes.

The probe paper conjectured:

Proposition 2.2 ([1], Prop. 7.1, conjectured)**.** *For every $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox T and satisfiable concept C_0, every weakly arc-consistent typed network with non-empty domains is realisable by some model of $T \cup \{C_0\}$.*

We now show Proposition 2.2 is false.

3 The counterexample

Example 3.1 (Two-node unrealisable network)**.** Let T be the following $\mathcal{ALCI}_{\mathsf{RCC5}}$ TBox with concept names A, B, D:

$$
\begin{aligned}
&(\mathrm{T1}) \quad A \sqsubseteq \exists \mathsf{PP}.D,\\
&(\mathrm{T2}) \quad A \sqsubseteq \neg B,\\
&(\mathrm{T3}) \quad B \sqsubseteq \forall \mathsf{DR}.\neg D,\\
&(\mathrm{T4}) \quad B \sqsubseteq \forall \mathsf{PO}.\neg D,\\
&(\mathrm{T5}) \quad B \sqsubseteq \forall \mathsf{PP}.\neg D,\\
&(\mathrm{T6}) \quad B \sqsubseteq \forall \mathsf{PPI}.\neg D,\\
&(\mathrm{T7}) \quad D \sqsubseteq \neg B.
\end{aligned}
$$

Take $C_0 = A$ and $V = \{x, z\}$ with

$$
\begin{aligned}
\tau(x) &\supseteq \{A,\ \neg B,\ \neg D,\ \exists \mathsf{PP}.D\},\\
\tau(z) &\supseteq \{B,\ \neg A,\ \neg D,\ \forall \mathsf{DR}.\neg D,\ \forall \mathsf{PO}.\neg D,\ \forall \mathsf{PP}.\neg D,\ \forall \mathsf{PPI}.\neg D\}.
\end{aligned}
$$

$\tau(x)$ and $\tau(z)$ are completed to Hintikka types for (C_0, T) by closure under TBox axioms and Boolean saturation; the completions are consistent and the choice of $\neg D \in \tau(x)$ is TBox-free (nothing in T forces D on an A-typed node).

Observation 3.2 (Arc-consistency)*.* The pair (V, D, τ) is weakly arc-consistent with non-empty domains, where $D(x, z) = \mathrm{TypeSafe}(\tau(x), \tau(z)) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$.

Proof. V has only two nodes and hence no triangles; composition propagation is vacuous. Arc-consistency reduces to the type-safety filter. We check $\mathrm{TypeSafe}(\tau(x), \tau(z))$ atom-by-atom.

- DR: $\tau(z)$ contains $\forall \mathsf{DR}.\neg D$, requiring DR-neighbours of z to have $\neg D$. $\tau(x)$ has $\neg D$. Safe. $\tau(x)$ has no DR-universal. Safe.
- PO: Analogous. $\tau(z)$'s $\forall \mathsf{PO}.\neg D$ is satisfied by $\tau(x)$'s $\neg D$. Safe.
- $\mathsf{PP}(x, z)$: equivalent to $\mathsf{PPI}(z, x)$. $\tau(z)$'s $\forall \mathsf{PPI}.\neg D$ requires x to have $\neg D$. Yes. Safe.
- $\mathsf{PPI}(x, z)$: equivalent to $\mathsf{PP}(z, x)$. $\tau(z)$'s $\forall \mathsf{PP}.\neg D$ requires x to have $\neg D$. Yes. Safe.
- EQ: forces $\tau(x) = \tau(z)$, but $A \in \tau(x)$ and $\neg A \in \tau(z)$ (from (T2)). Not safe.

Hence $\mathrm{TypeSafe}(\tau(x), \tau(z)) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, which is non-empty. No further propagation applies. □

Observation 3.3 (Unrealisability)*.* There is no model $M \models T$ containing elements x^M, z^M of the prescribed types.

Proof. Suppose M were such a model. From $\exists \mathsf{PP}.D \in \tau(x)$ there is $y \in M$ with $\mathsf{PP}(x, y)$ and $D \in \mathrm{tp}^M(y)$. By (T7), $\neg B \in \mathrm{tp}^M(y)$. Consider the atomic $\mathsf{RCC5}$ relation $R := \rho^M(y, z) \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$. We rule out every possibility:

- $R = \mathsf{DR}$: $\tau(z)$'s $\forall\mathsf{DR}.\neg D$ forces $\neg D \in \mathrm{tp}^M(y)$; but $D \in \mathrm{tp}^M(y)$. Contradiction.
- $R = \mathsf{PO}$: analogous, via $\forall\mathsf{PO}.\neg D$.
- $R = \mathsf{PP}$: i.e. $y\mathsf{PP}z$ equivalently $z\mathsf{PPI}y$. By (T5) applied at z (which has $\forall\mathsf{PP}.\neg D$): if $\mathsf{PPI}(z, y)$, then the PP-neighbour of z is y; wait: $\mathsf{PP}(y, z)$ means y is a proper part of z, equivalently z is the proper-containing relation PPI from y's view, and equivalently y is a proper part of z. From z's perspective, y is a PPI-neighbour (i.e. z's PPI-successor is y). $\tau(z)$'s $\forall\mathsf{PPI}.\neg D$ forces $\neg D \in \mathrm{tp}^M(y)$. But $D \in \mathrm{tp}^M(y)$. Contradiction.
- $R = \mathsf{PPI}$: dually, $z\mathsf{PP}y$, so y is a PP-neighbour of z; $\tau(z)$'s $\forall\mathsf{PP}.\neg D$ forces $\neg D \in \mathrm{tp}^M(y)$. Contradiction.
- $R = \mathsf{EQ}$: $y = z$, forcing $\mathrm{tp}^M(y) = \mathrm{tp}^M(z)$, so $\{B, D\} \subseteq \mathrm{tp}^M(y)$. But (T7) gives $D \sqsubseteq \neg B$, contradicting $B \in \mathrm{tp}^M(y)$.

No relation is possible. Hence no model. □

Corollary 3.4. *Proposition 2.2 is false: Example 3.1 is weakly arc-consistent with non-empty domains but has no model.*

4 What went wrong

The counterexample factors through a gap in Definition 2.1. Weak arc-consistency propagates constraints among *existing* nodes of V via composition and pairwise type-safety. It does not propagate the implicit constraint that every pending $\exists R.D \in \tau(x)$ must have, for each $z \in V$, some atomic $R' \in \mathrm{comp}(\mathrm{inv}(R), D(x, z))$ type-safe for a plausible witness type.

In Example 3.1, the pending existential $\exists\mathsf{PP}.D$ at x requires a witness y with $D \in \tau(y)$. For such a y and the existing z:

- $\mathrm{comp}(\mathsf{PPI}, D(x, z)) = \mathrm{comp}(\mathsf{PPI}, \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ (all atomic relations are in the composition).
- But $\mathrm{TypeSafe}(\tau(y), \tau(z))$ is empty for every $\tau(y)$ with D, by the same atom-by-atom argument as in Observation 3.3.

The *pending* filter is empty, but weak arc-consistency never computes the pending filter. The emptiness is invisible from within V.

Remark 4.1 (Where the probe paper's probes differ). The three probes of [1] (forced coincidence, inverse cycle, yardstick) all involved TBoxes where the obstructions were *between existing nodes* of the tried network. Forced coincidence put the universals between y and z both in V; inverse cycle tied up three or four existing nodes; yardstick forced type-saturation before any network was constructed. None of the probes was designed to probe the existential-witness extensibility gap specifically. That gap requires the obstruction to sit on an edge where one endpoint is inside V and the other is a pending witness outside V. Example 3.1 does exactly this.

5 The fix: witness-menu-style propagation

The minimal correction is to augment Definition 2.1 with an existential-propagation clause.

Definition 5.1 (Strong arc-consistency)**.** (V, D, τ) is *strongly arc-consistent* if it is weakly arc-consistent (Def. 2.1) *and* additionally: for every $x \in V$ and every $\exists R.D \in \tau(x)$, there exists a Hintikka type $\gamma \in \mathrm{Tp}(C_0, T)$ with $D \in \gamma$ and the forward-propagation $\{E : \forall R.E \in \tau(x)\} \subseteq \gamma$, such that for every $z \in V \setminus \{x\}$:

$$\mathrm{comp}(\mathrm{inv}(R), D(x, z)) \cap \mathrm{TypeSafe}(\gamma, \tau(z)) \neq \varnothing.$$

The new clause says: every pending existential has a candidate witness type whose atomic connection to every existing node is non-empty. This is exactly the information a *witness menu* in the split-forest paper's rank-d quotient framework [2] provides: a target type γ and a set of permitted relations $\mathrm{comp}(\mathrm{inv}(R), D(x, z)) \cap \mathrm{TypeSafe}(\gamma, \tau(z))$ for each existing node.

Proposition 5.2 (Fix; not proven here)**.** *Strong arc-consistency is complete for realisability of typed $\mathcal{ALCI}_{\mathsf{RCC5}}$ networks.*

We do not prove Proposition 5.2 in this paper. A proof would repackage the split-forest soundness argument (König's lemma on canonical refinements, LCA induction) with strong arc-consistency replacing the more elaborate rank-d witness-menu propagation. We conjecture the two are equivalent modulo bookkeeping, but this is not trivial; the split-forest paper's rank-d hierarchy handles several additional concerns (EQ-sync, sibling-interface descriptor table coherence (C1)–(C5)) that strong arc-consistency in its simplest form does not.

In particular: Example 3.1 is caught by strong arc-consistency. For $\exists \mathsf{PP}.D$ at x: no Hintikka type γ containing D has non-empty $\mathrm{TypeSafe}(\gamma, \tau(z))$. The existential clause of Definition 5.1 fails, and the network is flagged as unrealisable.

6 Implications

6.1 For decidability

Nothing changes. The split-forest soundness theorem [2] and the completeness extraction theorem [3] remain intact. Both use propagation stronger than weak arc-consistency; neither relied on Proposition 2.2. Decidability of $\mathcal{ALCI}_{\mathsf{RCC5}}$ is unaffected.

6.2 For a "simpler alternative proof"

The probe paper closed with the sketch of an ambitious theorem: "every $\mathcal{ALCI}_{\mathsf{RCC5}}$ constraint on a finite network has arity ≤ 3, and arc-consistency is complete for arity-≤ 3 CSPs over $\mathsf{RCC5}$" ([1], §9). The counterexample here refutes the naive version of this claim.

Existentials in ALCI are *non-local* constraints: they force the existence of additional nodes whose relations to the entire current network must be simultaneously consistent with the TBox. Packing this into a CSP of fixed arity requires either (a) an unbounded-arity encoding, or (b)

explicit existential variables with domain $\mathrm{Tp}(C_0, T)$ and complex binary constraints between them and each $z \in V$. Option (b) is essentially what witness menus in the split-forest paper are.

There is no simpler alternative decidability proof of the form sketched. Witness-menu-style existential propagation is the minimum viable structure.

6.3 For the undecidability landscape

The probe paper observed that if typed patchwork held, every known undecidability reduction would be blocked by a single structural fact (arity-≤ 3 CSP completeness). Now that the naive typed patchwork is false, that particular narrative does not go through. However, the undecidability picture is not reopened: the correct narrative is merely more nuanced.

- Strong arc-consistency (Def. 5.1) still has the "no k-ary synthesis for $k \geq 4$" property as argued in [1], §8. Existential clauses add binary constraints between the pending witness type γ and each $z \in V$, not genuine k-ary constraints.
- Strong arc-consistency is what the split-forest soundness uses, and the soundness is proven. So strong arc-consistency *is* complete for quotient-derived networks, which suffice for decidability.
- The structural reason no undecidability reduction has succeeded remains: ALCI constructs decompose into node-local, arc-local, triple-local, and existential-extension constraints, all of which are captured by strong arc-consistency + rank-d bookkeeping.

The shift is: the "load-bearing" place in the decidability proof is not "k-ary CSP completeness" (which would be false) but "strong arc-consistency with explicit existential propagation is complete for quotient-derived typed networks" (which is what the split-forest paper actually proves).

7 Honest assessment

This paper is a focused correction to a specific conjecture (Proposition 2.2). What we establish:

1. A concrete two-node, seven-axiom counterexample (Example 3.1) that is weakly arc-consistent with non-empty domains and has no model.
2. The counterexample factors through the gap between intra-V type-safety (which weak arc-consistency sees) and witness-extensibility (which it does not).
3. The minimal fix (Definition 5.1) adds a witness-menu-style propagation clause and is conjectured to be complete (Proposition 5.2), though we do not prove it.
4. The split-forest paper's apparent complexity (rank-d witness menus, canonical refinements) is load-bearing: witness extensibility must be propagated explicitly.

What we do not establish:

1. A proof of Proposition 5.2. The substance of that proof lies in the split-forest paper; formally deriving strong arc-consistency completeness in a simpler presentation would be worthwhile but non-trivial.

2. Any change to the decidability picture for $\mathcal{ALCI}_{\mathsf{RCC5}}$. The probe was aimed at undecidability (via a typed-patchwork failure); this counterexample is to a proof sketch, not to the decidability itself.

3. A sharper CSP-theoretic framing. The arity-$\leq$ 3 sketch in [1], §9 would need to be reformulated to account for existential extension, and we do not attempt this reformulation here.

Methodological note. The probe paper's three probes were carefully structured but all *within a fixed* V. The counterexample here sits on the boundary between V and its pending-witness extension—structurally different from anything the probes tested. This is the methodological lesson: when designing adversarial probes against a conjecture, the search space must include obstructions that cross the boundary between the stated network and its implicit extensions. Obstructions confined to existing nodes are not representative. This is the same "distribution blindness" that Opus 4.7's adversarial review surfaced in the cover-tree tableau [5]: tests designed within the generator's distribution miss bugs outside it. A similar principle applies to conjectures.

References

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